



Bansilal Ramnath Agarwal Charitable Trust's
Vishwakarma Institute of Information Technology

**Department of
Artificial Intelligence and Data Science**

Student Name: Sathe Onkar Abasaheb

Class: TY

Division: B

Roll No: 372078

Semester: V

Academic Year: 2025-2026

Subject Name & Code: Cloud and DevOps

Assignment No: 01

Title of Assignment: Execute basic Linux commands (file operations, process management, user management)

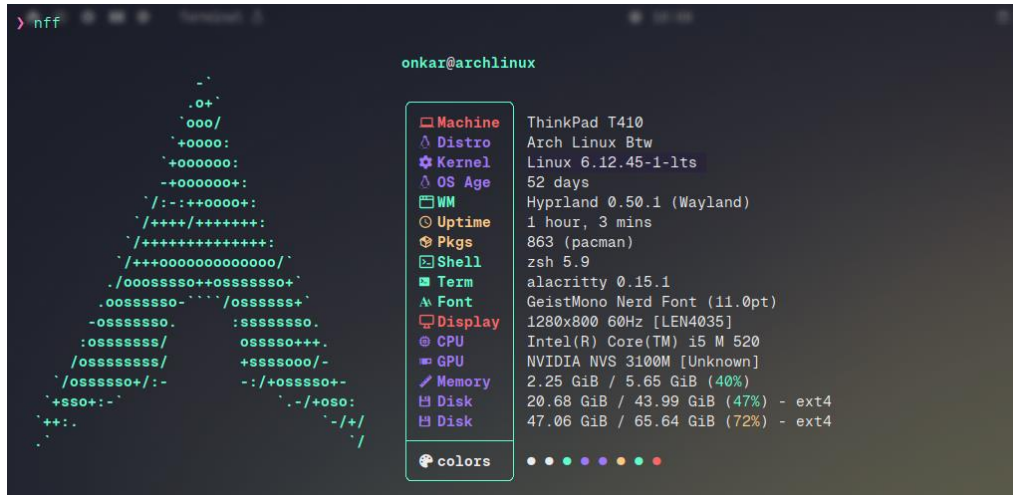
Date of Performance: / /2025

Date of Submission: / /2025

Aim: To execute Linux commands (file operations, process management, user management)

Objective: To learn basic Linux commands like file operations, process management and user management

Materials Required: Laptop with Arch Linux(OS kernel Linux 6.12.45-1-lts) installed in it, zsh(highly configured .zshrc).



The image shows a terminal window with a dark background. On the left, there is a large ASCII art tree structure. On the right, there is a box containing system information. The terminal prompt is 'onkar@archlinux'.

```
onkar@archlinux
```

Machine	ThinkPad T410
Distro	Arch Linux Btw
Kernel	Linux 6.12.45-1-lts
OS Age	52 days
WM	Hyprland 0.50.1 (Wayland)
Uptime	1 hour, 3 mins
Pkgs	863 (pacman)
Shell	zsh 5.9
Term	alacritty 0.15.1
Font	GeistMono Nerd Font (11.0pt)
Display	1280x800 60Hz [LEN4035]
CPU	Intel(R) Core(TM) i5 M 520
GPU	NVIDIA NVS 3100M [Unknown]
Memory	2.25 GiB / 5.65 GiB (40%)
Disk	20.68 GiB / 43.99 GiB (47%) - ext4
Disk	47.06 GiB / 65.64 GiB (72%) - ext4

colors

Procedure:

File Operations: File Operations are tasks performed on files and directories such as creating, copying, moving, and deleting.

□ Commands used:

- o touch – Create a file
- o cp – Copy files/directories
- o mv – Move or rename
- o rm – Remove
- o ls – List contents
- o cat, more, less – View file contents
- o chmod, chown – Manage permissions and ownership

Purpose:

To organize data, manage storage, and control access.

Step 1: Open terminal

Step 2: Create a folder

Step 3: Create a new text file

Step 4: Rename the text file

Step 5: Write text in the file

Step 6: Delete the file and folder

```

Terminal
12-Sep 180% 40%
> ps
  PID TTY          TIME CMD
 14289 pts/0    00:00:01 zsh
 143522 pts/0    00:00:00 zsh
 143527 pts/0    00:00:00 gitstatusd-linu
 145921 pts/0    00:00:00 ps

> who
onkar    tty2            2025-09-12 21:05
onkar    pts/1          2025-09-12 21:13 (tmux(19195).%0)
onkar    pts/0          2025-09-12 22:10 (tmux(19195).%1)
onkar    pts/3          2025-09-12 22:10 (tmux(19195).%2)
onkar    pts/4          2025-09-12 22:10 (tmux(19195).%3)

~ >
10:10:34 PM

> mkdir Temp$$
> ls
DATABASE      Lienzo-Backend      recovery
db             'Lienzo-Backend (2)' SIH25
DB             'Lienzo-Backend (3)' Temp145250
Desktop        'Lienzo-Backend (4)' test
Documents      'Lienzo-Backend (5)' Videos
dotfiles       mongo_data
Downloads      Music
DSA            node_modules
FastShop       Pictures
frontend-better-auth Practical
git-clones     Practice
inbox          Projects
Learn          react-playground

~ >
10:10:51 PM

cat server.js
File: server.js
1  require("dotenv").config();
2  const express = require("express");
3  const ConnectMongo = require("../config/conn/db.js");
4  const app = express();
5
6  // Connect to database
7  ConnectMongo();
8
9  // Setup all middleware
10 const setupMiddleware = require("../api/middleware/index.
    middleware.js");
11 setupMiddleware(app);
12
13 // Routes
14 const products = require("../api/routes/product.routes.js
    ");
15 const home = require("../api/routes/home.routes.js");
16
17 app.use("/products", products);
18 app.use("/", home);
19
20 app.listen(process.env.PORT, () => {
21   console.log( Listening on port: ${process.env.PORT}
22 );
23 });

~/FastShop/Backend main* >
10:11:18 PM
UP: 1h 5m
```

[illegible]

User Management:

User Management refers to creating, modifying, and deleting user accounts in an operating system. It ensures system security and resource access control.

▢ Commands used:

- o **adduser** / **useradd** – Add a new user
- o **passwd** – Set or change user password
- o **deluser** / **userdel** – Delete a user
- o **usermod** – Modify a user account

▢ Groups are used to manage permissions for multiple users together. Purpose:

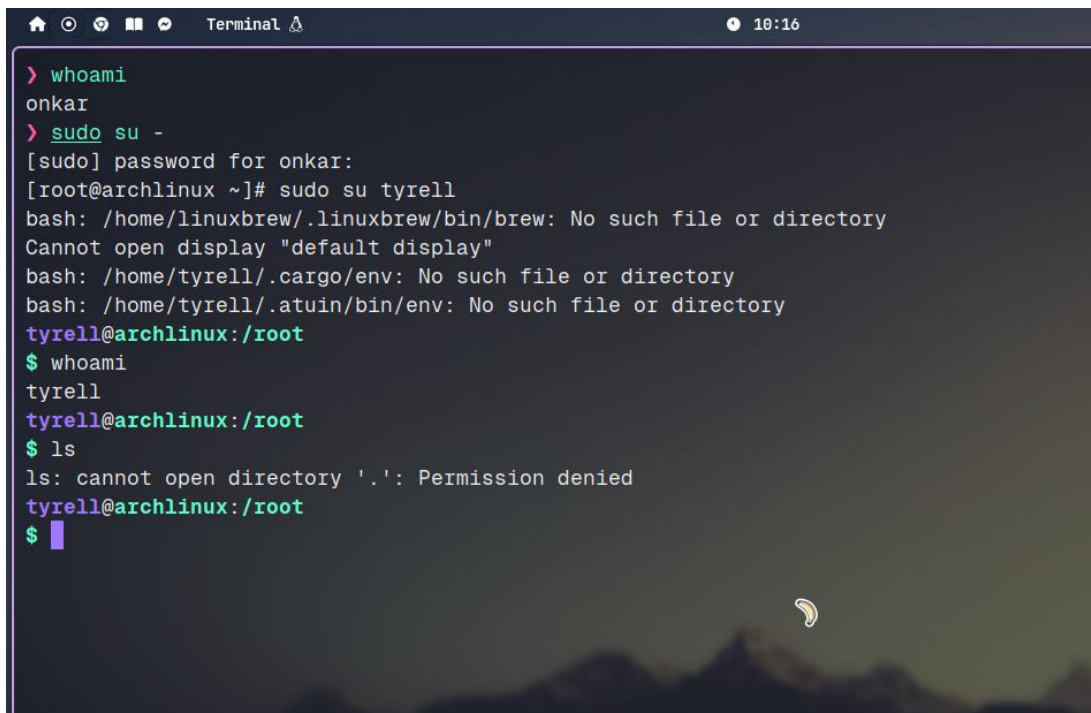
To provide personalized environments and restrict access based on roles.

Step 1: Open Terminal

Step 2: Command to display username

Step 3: Command to add a user

Step 4: Command to add new password of the user



```
> whoami
onkar
> sudo su -
[sudo] password for onkar:
[root@archlinux ~]# sudo su tyrell
bash: /home/linuxbrew/.linuxbrew/bin/brew: No such file or directory
Cannot open display "default display"
bash: /home/tyrell/.cargo/env: No such file or directory
bash: /home/tyrell/.atuin/bin/env: No such file or directory
tyrell@archlinux:/root
$ whoami
tyrell
tyrell@archlinux:/root
$ ls
ls: cannot open directory '.': Permission denied
tyrell@archlinux:/root
$
```

Permission Management Commands:

- chmod - Change file permissions
- chown - Change file ownership
- chgrp - Change group ownership
- umask - Default permission mask

✦ Text Processing & Search

cat - Concatenate and display files
less - View file content one screen at a time
head - Show first lines of file
tail - Show last lines of file
grep - Search text using patterns
find - Search for files
locate - Find files (fast, indexed)

✦ Process Management

ps - Show running processes
top - Real-time system monitor
htop - Interactive process viewer
kill - Terminate process by PID
killall - Kill process by name
bg - Resume background job
fg - Resume foreground job
jobs - List active shell jobs
nice - Start process with priority
renice - Change priority of running process

✦ Networking

ping - Send ICMP echo request
ip addr - Show IP addresses
ifconfig - Configure network interfaces (older)
netstat - Show network status (deprecated)
ss - Show sockets (modern replacement)
curl - Transfer data via HTTP

wget - Download files from web

scp - Secure file copy over SSH

ssh - Secure shell access

ftp/sftp - File transfer protocols

traceroute - Trace network route

dig/nslookup - DNS queries

hostname - Show or set hostname

Conclusion:

Though this assignment I explored the Linux OS(Arch BTW), where I explored to manage everything from cursor to window manager and config files of each individual application I am using(Neovim, zsh, tmux and etc)
Along with that I learned to manage users, basic commands and etc.